Capgemini Exceller Hiring Process

Answer Key & Explanations

Section A: AI Literacy

Q1. A - LLM = Large Language Model, trained on huge text data to understand and generate language.
Q2. B - Hallucination = fluent but false/fabricated output. Always verify.
Q3. B - The prompt is what you give the model.
Q4. C - Clear language, task, constraints and tests give better output.
Q5. A - Zero-shot = task without examples.
Q6. B - Few-shot = a few examples inside the prompt.
Q7. B - Models read and generate text as tokens.
Q8. B - Biased or unbalanced training data leads to biased outputs.
Q9. B - Always review, understand and test AI code.
Q10. A - High temperature = more varied/random outputs; low = more deterministic.
Q11. C - A fixed formula is deterministic; it does not generate new content.
Q12. B - Protect confidential data; use approved tools only.
Q13. B - It is the skill of crafting effective prompts.
Q14. A - Step-by-step reasoning often improves complex answers.
Q15. B - It is the model's working memory limit.
Q16. B - RAG retrieves relevant data and feeds it to the model for grounded answers.
Q17. B - Responsible AI = fairness, transparency, accountability, privacy, safety.
Q18. B - Verify and iterate.
Q19. A - Labelled data = supervised learning.
Q20. B - Fine-tuning adapts a pretrained model to a task or domain.

Section B: Technical - DSA, Pseudocode, CS Fundamentals

Q21. B - Search space halves every step.
Q22. B - Last In First Out.
Q23. B - First In First Out.
Q24. B - x=8; y=8-3=5; x=8-5=3. Output 3 5 (swap).
Q25. B - 1+2+3+4+5 = 15.
Q26. B - 10>5>2>1>0 takes 4 divisions.
Q27. C - 5! = 120.
Q28. B - 0,1,1,2,3,5,8 -> f(6)=8.
Q29. C - a[0]=2, so a[2]=6.
Q30. B - 101 & 011 = 001 = 1.
Q31. C - 101 ^ 011 = 110 = 6.

Q32. C - Left shift by 4 = 1 x 2^4 = 16.

Q33. C - Nested passes give O(n^2).

Q34. B - Merge sort splits, sorts and merges: O(n log n) in all cases.

Q35. B - BFS uses a queue.

Q36. B - DFS uses a stack (or recursion).

Q37. B - Primary key = unique + not null.

Q38. C - DROP TABLE removes data and structure.

Q39. B - Atomicity: all or nothing.

Q40. C - The condition is "no preemption", so preemption is not a condition.

Q41. C - Network layer (IP routing).

Q42. B - Encapsulation.

Q43. C - 1+2+3 = 6.

Q44. B - 17 % 5 = 2; 17 // 5 = 3; total 5.

Q45. A - Only pointer changes: O(1).

Section C: Debugging Assessment

Q46. Bug: i < n skips n. Fix: i <= n.

Q47. Division by zero when list is empty (ZeroDivisionError). Fix: if not a: return 0 (or raise a clear error).

Q48. == compares references, not content. Fix: if (s1.equals(s2)).

Q49. Assignment (=) used instead of comparison (==); x becomes 5 (true). Fix: if (x == 5).

Q50. Missing increment. Add i += 1 inside the loop.

Q51. Loop goes to i = 5 which is out of bounds (valid 0-4). Fix: i < 5.

Q52. Mutable default argument is shared across calls. Fix: def add(x, lst=None): if lst is None: lst = []

Q53. Integer division happens before conversion to double. Fix: (a + b) / 2.0.

Q54. Missing break (fall-through). Add break; after each case.

Q55. No base case. Add: if n <= 1: return 1.

Section D: AI-Assisted Coding

Q56. Include: language (Python), function name/signature, input and output type, rules (ignore case/spaces), edge cases (empty string, single char), constraints (no external libs), and ask for 3-4 test cases. Example: "Write a Python function is_palindrome(s) returning True/False, ignoring case and spaces; handle empty string; no libraries; include 4 test cases."

Q57. Run it, copy the failing test and error, tell the AI exactly what fails (input, expected, actual), ask for a fix, then re-run all tests including edge cases. Do not accept the first answer blindly.

Q58. State constraints clearly: "n up to 10^5, need O(n log n) or better; explain approach and complexity." Then verify the new code against a small brute-force version.

Q59. 1) Understood every line. 2) Ran provided + own tests. 3) Edge cases (empty, one element, max size, negatives). 4) Checked time/space complexity. 5) Checked input/output format. 6) Cleaned unused code.

Q60. Break into steps: (1) define data structure and functions; (2) implement add with validation; (3) implement search; (4) implement delete; (5) menu loop; (6) tests. Review and test after each prompt, not only at the end.

Section E: English Communication

Q61. B - Third-person singular present: goes.

Q62. C - "Neither" takes a singular verb.

Q63. B - Rapid = quick.

Q64. B - Scarce vs abundant.

Q65. C - Accommodation (double c, double m).

Q66. B - "Since" is used with a point in time.

Q67. C - Good at.

Q68. C - Second conditional uses "were".

Q69. B - Idiom: ease initial awkwardness.

Q70. B - Object becomes subject + was/were + past participle.

Section F: Cognitive / Game-Based Practice

Q71. C - Differences 4,6,8,10,12 -> 30+12 = 42.

Q72. C - Each term is doubled.

Q73. D - Cube is 3D; others are 2D shapes.

Q74. B - 4+15+7 = 26.

Q75. B - Next clockwise position is Left.

Q76. B - Row2 col3 = Circle; Row3 = Triangle, Circle, Square.

Q77. C - A > B > C.

Q78. C - 3 hours x 30 = 90 degrees.

Q79. B - Reversing gives 4 9 2 7.

Q80. C - Squares of 1 to 6.

Section G: Behavioral / PowerSkills Situations

Q81. Talk to them privately first, understand the reason, offer help, divide tasks clearly; involve the mentor only if the issue continues.

Q82. Ask clarifying questions early, list assumptions, prioritise core features, share progress updates, and deliver in small tested parts.

Q83. Own it quickly, inform the team, fix or propose a fix, and put a check in place so it does not repeat.

Q84. Listen first, compare options using facts (time, risk, quality), propose a small trial if possible, and support the final team decision.

Q85. Set a plan, learn basics from official docs/tutorials, build a small project, ask for feedback, and revise daily.

Section H: Technical Interview

Q86. Structure: problem > your role > tech stack > architecture > challenge and how you solved it > result. Be ready to explain every module and your own contribution.

Q87. Array: contiguous memory, fast index access O(1), costly insert/delete. Linked list: nodes with pointers, easy insert/delete, access O(n).

Q88. Encapsulation, Abstraction, Inheritance, Polymorphism, each with a short real example.

Q89. Process: independent program with its own memory. Thread: lightweight unit inside a process sharing memory; faster context switch.

Q90. Organising tables to reduce redundancy and anomalies (1NF, 2NF, 3NF, BCNF) by splitting data and using keys.

Q91. Stack: automatic, fast, function calls and local variables, LIFO. Heap: dynamic allocation, manual/garbage-collected, slower.

Q92. API is an interface for software to communicate. REST uses HTTP methods (GET, POST, PUT, DELETE) on resources, usually returning JSON.

Q93. DELETE removes selected rows (can use WHERE, rollback possible). TRUNCATE removes all rows, keeps structure. DROP removes the table and structure.

Q94. Use for ideas and boilerplate, give clear prompts, never share confidential data, review and test all output, understand the code, and give credit where required.

Q95. Works on a sorted array: compare with middle, discard half each step; O(log n). Prerequisite: data must be sorted.

Section I: HR Interview

Q96. 60-90 seconds: education, key skills/projects, one achievement, and interest in the role. Present > past > future.

Q97. Mention global consulting/IT services, learning culture, AI and digital focus, and growth with real projects. Link it to your skills.

Q98. Give 2 relevant strengths with proof; give a real, minor weakness plus what you are doing to improve it.

Q99. Be honest. A flexible, positive answer showing readiness to learn and adapt works best.

Q100. Growing technical skills, taking more responsibility, contributing to projects and moving toward a lead role at the company.